National Teachers College
629 J Nepomuceno, Quiapo, Manila, 1001 Metro Manila
Bachelor of Science in Information Technology

Week 12: MILESTONE 2 (MIDTERMS) — Dart Logic Practical

Mobile Development

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Design Reflection Paragraph

For the purpose of implementing this Dart logic program, my focus was on the key aspects of scalability, robustness, and good coding practices that would resemble the industry standards for mobile development. In order to achieve this, I applied an object-oriented design by creating an abstract base class (BaseEntity) and its concrete implementation (DataModel) to impose strict data contracts and ensure modularity and code reusability. For handling any manipulations with the data, I created a separate service class (ProcessingService) to follow the principle of single responsibility. Strict typing, null safety, and named constructor arguments were used intentionally to avoid any potential runtime errors and improve code readability. Moreover, asynchronous operations were added through Future and async/await to imitate actual communication with APIs and ensure flawless execution without any thread blocking. In accordance with the TANGLAW principles of practical competence and continuous learning, the mentioned decision shows how good logic and architecture directly affect the creation of professional mobile application backends.